Homework 4

How to Make AI in Law and Healthcare (More) Trustworthy?

Assignment Description

In this assignment, you will explore how AI systems can retrieve and provide factual information in high-stakes domains such as law and healthcare. You will build and test a Retrieval-Augmented Generation (RAG) chatbot that answers sensitive questions using real documents, such as health guidance or legal rights resources.

Your task is to compare two versions of an AI assistant:

- **Base Model**: answers from its internal Language Model knowledge only.
- **RAG Model**: retrieves relevant reference documents before answering.

You will run both models on the same set of questions, analyze their differences, and reflect on how factual grounding changes the AI responses. You'll also explore how techniques like embeddings, similarity search, and fine-tuning influence behavior in real-world contexts. The goal is not to build a polished system, but to **make the invisible design choices visible**, like how data access, retrieval, and safeguards shape the reliability of AI assistants. Finally, you will reflect on what it would mean for organizations to deploy systems like these responsibly.

Deliverables

You are expected to **submit two items** for this assignment on Canvas:

1. The PDF of your Google Colab Notebook
2. The file you add to your RAG database

Database for RAG

You will need to download this ZIP file that contains the data you will use for this assignment.

Accessing the Assignment Through Google Colab

The notebook you will use for this assignment is here. **To get started, create your own copy of the notebook in your personal drive.** You can do this by:

Clicking "File" → Clicking "Save a copy in Drive"

This copy will be the file you will edit.